

MOHAMMED RUPAWALA

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Education

Adani University

Ahmedabad, Gujarat, India

B.Tech in Information and Communication Technology

Aug 2022 – Present

- CGPA: **7/10**
- Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Deep Learning, Computer Vision

Technical Skills

Languages: JavaScript, TypeScript, Python, C++, SQL, GO, HTML/CSS

Technologies: Node JS, FastAPI, React JS, NextJS, Tailwind

Tools: Vs Code, Docker, Git, GitHub

Experience

E2M Solutions

Jan 2026 – Present

AI Engineering Intern

- Designed and launched a multi-tenant SaaS AI chatbot platform used by 20+ client organizations, enabling plug-and-play knowledge base ingestion (PDFs, docs, URLs).
- Built end-to-end ingestion and retrieval pipelines using embeddings and semantic search, reducing client support workload by 40–60% and improving response accuracy across domains.
- Integrated APIs and workflow orchestration to automate business processes.

Projects

CUTI | *GO, Linux*

Feb 2026

- Built a single-threaded, event-driven TCP server in Go using low-level `syscall` and `epoll`, handling 10k+ concurrent connections with non-blocking I/O and $O(1)$ event dispatch, mirroring Redis' core architecture.
- Implemented the Redis Serialization Protocol (RESP) from scratch, supporting Arrays, Bulk Strings, and Integers, enabling command pipelining and improving request throughput by 3–4×.
- Designed a probabilistic key-expiration algorithm inspired by Redis active expiry cycle and the CLT.
- Implemented Append-Only File (AOF) persistence with atomic background rewrites, simulating Copy-on-Write semantics to ensure crash-safe recovery while maintaining read/write availability.

Deki | *Go, OS, CLI*

Jan 2026

- Built a lightweight Unix-like shell in Go supporting command parsing, execution, and built-in commands.
- Implemented CLI command autocomplete using a Trie data structure for fast prefix-based suggestions.
- Implemented process management and command execution using Go's `os/exec` and system-level abstractions.

MindSync | *React, Tailwind, FastAPI, TypeScript, PostgreSQL, Redis, LangChain*

Apr 2025

- Built an AI-powered journaling platform using RAG and vector embeddings to enable context-aware search and conversational querying across long-form personal journals.
- Integrated a Hugging Face-based sentiment analysis model to perform multi-class mood classification on journal entries of up to 1,000+ words, extracting actionable emotional insights.
- Optimized the retrieval and inference pipeline to parse, analyze, and respond to user queries within 7 seconds end-to-end, balancing accuracy and latency for real-world usage.

Hackathons

Hackspire: Finalist (Top 5), Adani University

Apr 2025

Built a smart queue management system that predicts optimal user arrival times using demand forecasting, reducing wait times and improving service efficiency.

Odoo Hackathon: Final Round, GIFT City, Gujarat

Jul 2025

Developed a Q&A platform with features like question tagging, threaded answers, and voting, enabling efficient knowledge sharing within organizations.

Competitive Programming

LeetCode(THFC): 1650 — Codeforces(Special_One): 1200